Design of Artificial Intelligence Driven Products

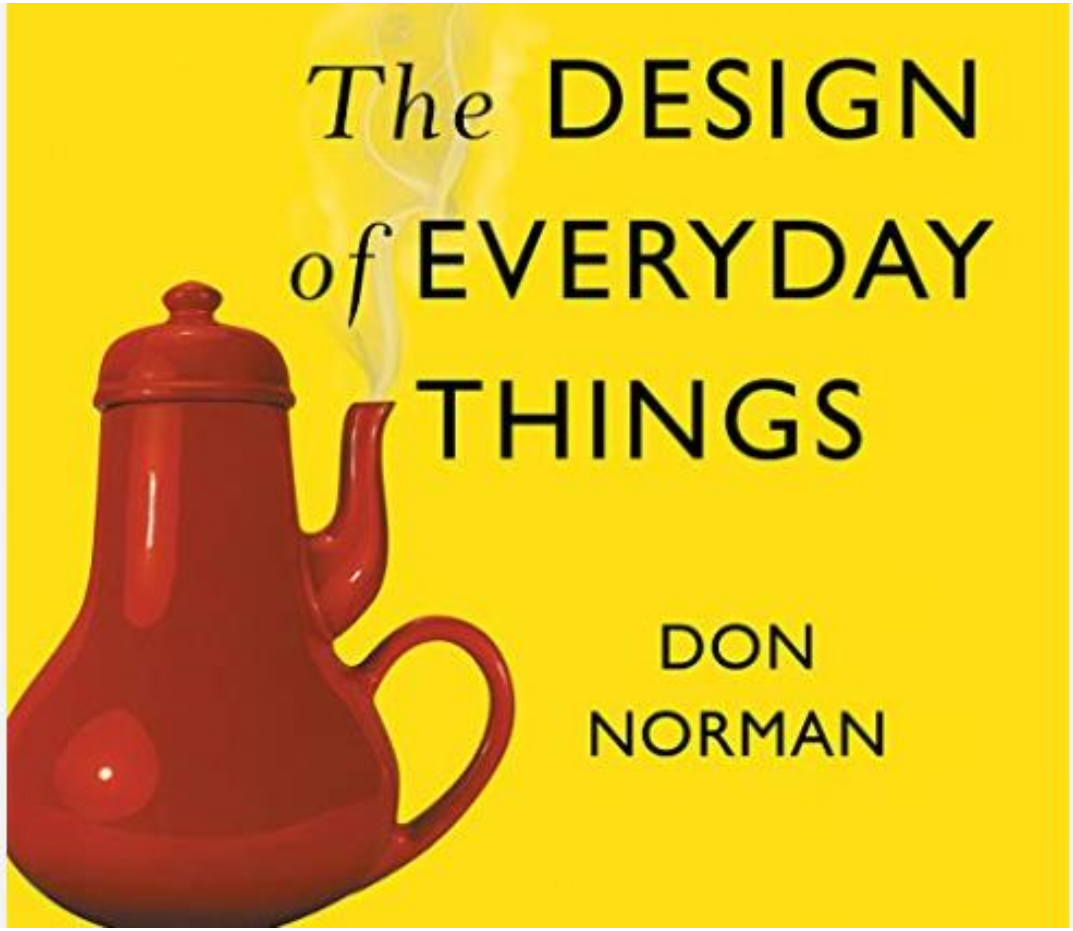
DES 529

IIIT Delhi

January – April 2024

Dr. Vinish Kathuria

Human Centered Design



Ch 1 – The Psychopathology of EveryDay Things

- Two most important characteristics of good design are Discoverability and Understanding

- Your project design should keep this in mind even for an AI-driven digital solution - Discoverability and Understanding

Linkage of Ethics to Design

- Two main pillars of good corporate governance are Accountability and Transparency

- Discoverability and Understanding as Design Pillars – Can you have Understanding without Transparency and Accountability?

Ethics – try to get answers for your project

What ethics should I keep in mind while building an AI product to help patients with Alzheimer's

1. **Transparency and Explainability**: Make AI models interpretable and avoid black box approaches.
2. **Privacy and Data Security**: Obtain informed consent, anonymize data, and ensure secure storage.
3. **Bias and Fairness**: Regularly assess for bias and train on diverse data.
4. **Clinical Validation and Safety**: Conduct trials, monitor adverse effects, and ensure robustness.
5. **Collaboration with Healthcare Professionals**: Involve experts and prioritize user-centered design.
6. **Long-Term Support and Maintenance**: Commit to continual improvement and ethical oversight.

Ref – Chat with Co-pilot 28/03/24

Linkage of Ethics to Design

Dig deeper in any points When designing an AI product for patients with Alzheimer's disease (AD), achieving transparency and explainability is crucial. Here are some steps to ensure transparency and enhance user trust:

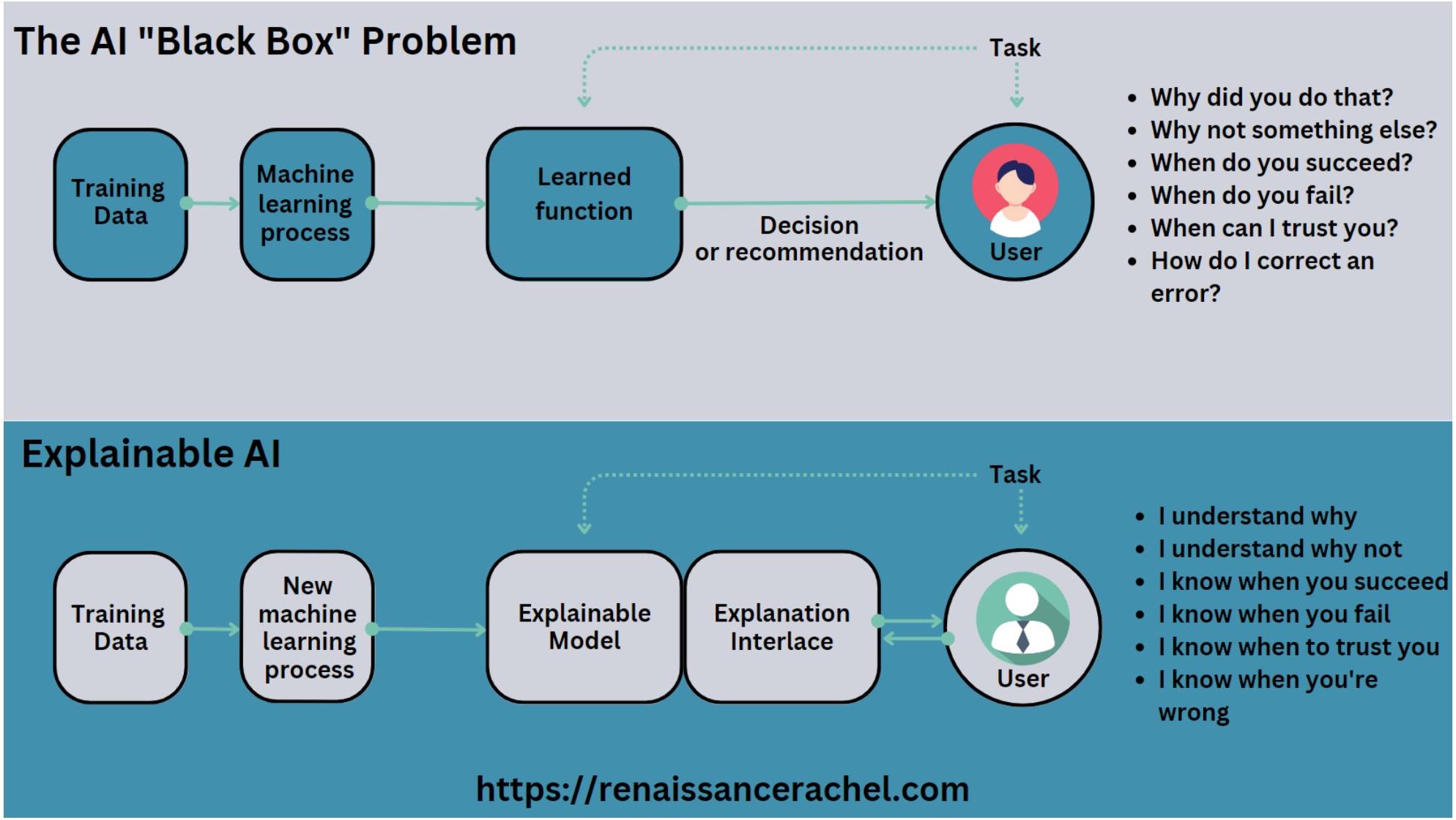
- **Feature Selection and Preprocessing**:
  - Choose relevant features related to Alzheimer's diagnosis, such as cognitive tests, brain imaging data, or patient history.
  - Preprocess data to handle missing values, outliers, and ensure consistency.
- **Interpretable Model Selection**:
  - Opt for models that are inherently interpretable, such as decision trees, linear regression, or rule-based systems.
  - Avoid complex models like deep neural networks unless you can provide clear explanations.
- **Feature Importance Analysis**:
  - Calculate feature importances to understand which features contribute most to predictions.
  - Techniques like SHAP values or permutation importance can help.

Ref – Chat with Co-pilot 28/03/24

Linkage of Ethics to Design

- **Local Explanations:**
  - Explain individual predictions for specific patients.
  - For instance, use LIME to highlight which features influenced a particular diagnosis.
- **Visualizations:**
  - Create visualizations to show how the model processes input data.
  - Display decision boundaries, feature interactions, or activation maps from neural networks.
- **Patient-Centric Explanations**:
  - Translate model outputs into patient-friendly language.
  - Explain why a certain diagnosis was made based on specific symptoms or test results.
- **Feedback Loop with Clinicians:**
  - Collaborate with healthcare professionals to validate model decisions.
  - Gather feedback on model behavior and explanations.
- **Documentation and Reporting**:
  - Document the model's architecture, training process, and evaluation metrics.
  - Provide a clear explanation of the model's purpose and limitations.

Explainable AI



Explainable AI

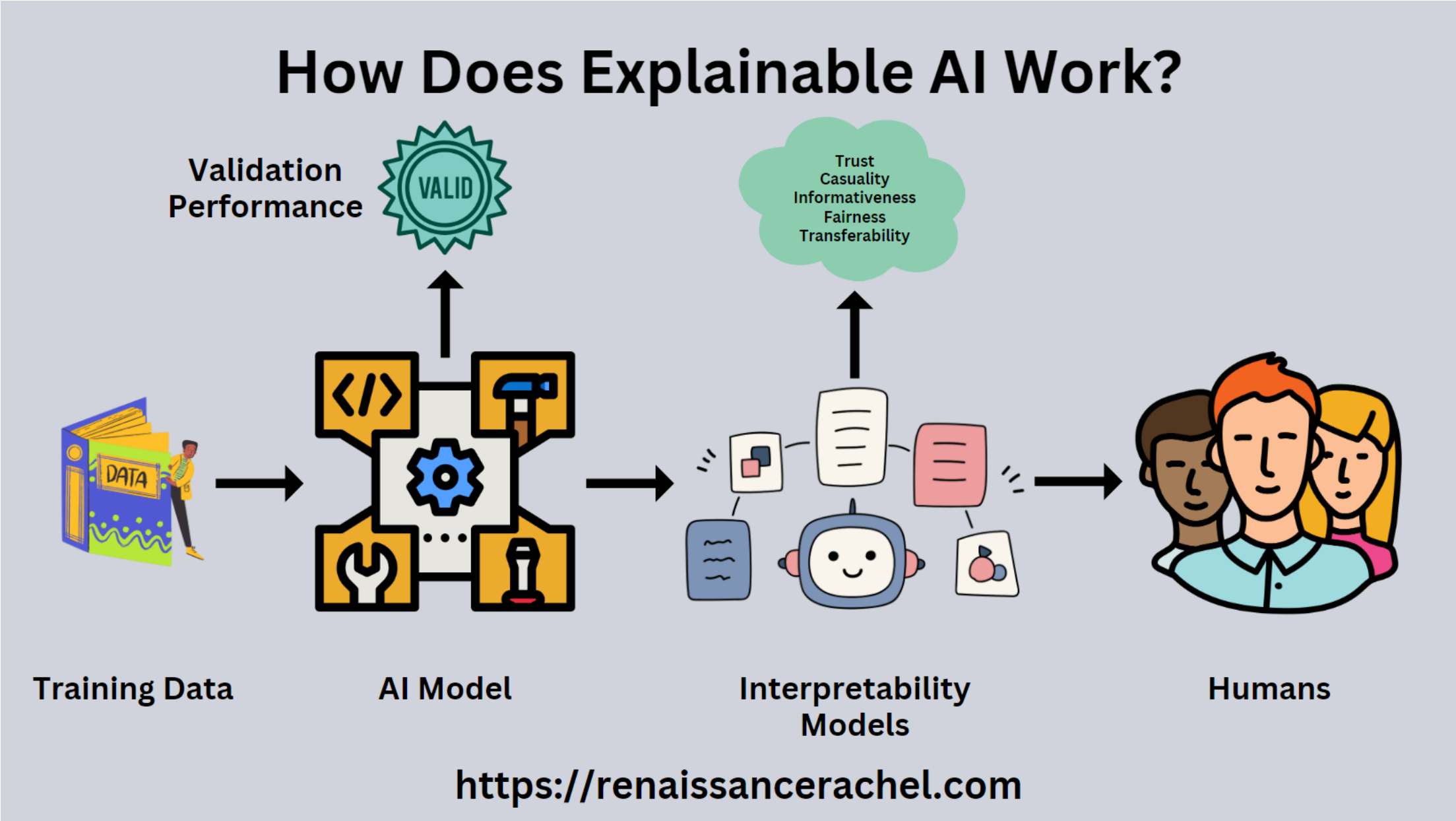
- Explainable AI is a form of AI designed to explain why it made its own decisions. Instead of just an output, XAI provides interactive explanations of how it works, and it comes to its conclusions. This helps AI practitioners understand the rationale behind AI systems and boosts trust in their decisions.

- **An XAI model uses natural language processing and knowledge graphs to explain how decisions were made, which are used to compare model predictions and performance in various fields, from healthcare to finance.** XAI algorithms are combined with traditional AI models to uncover the logic; this is an essential step towards responsible AI, which helps increase artificial intelligence transparency.

- So, what does all this mean for you? XAI can create a more transparent and accountable world, and you can trust your AI systems to make fair and responsible decisions. Transparency makes it easier for organizations to introduce autonomous systems into their operations as they can explain the rationale behind their choices. It also allows individuals to confidently use AI tools, knowing the system is trustworthy.

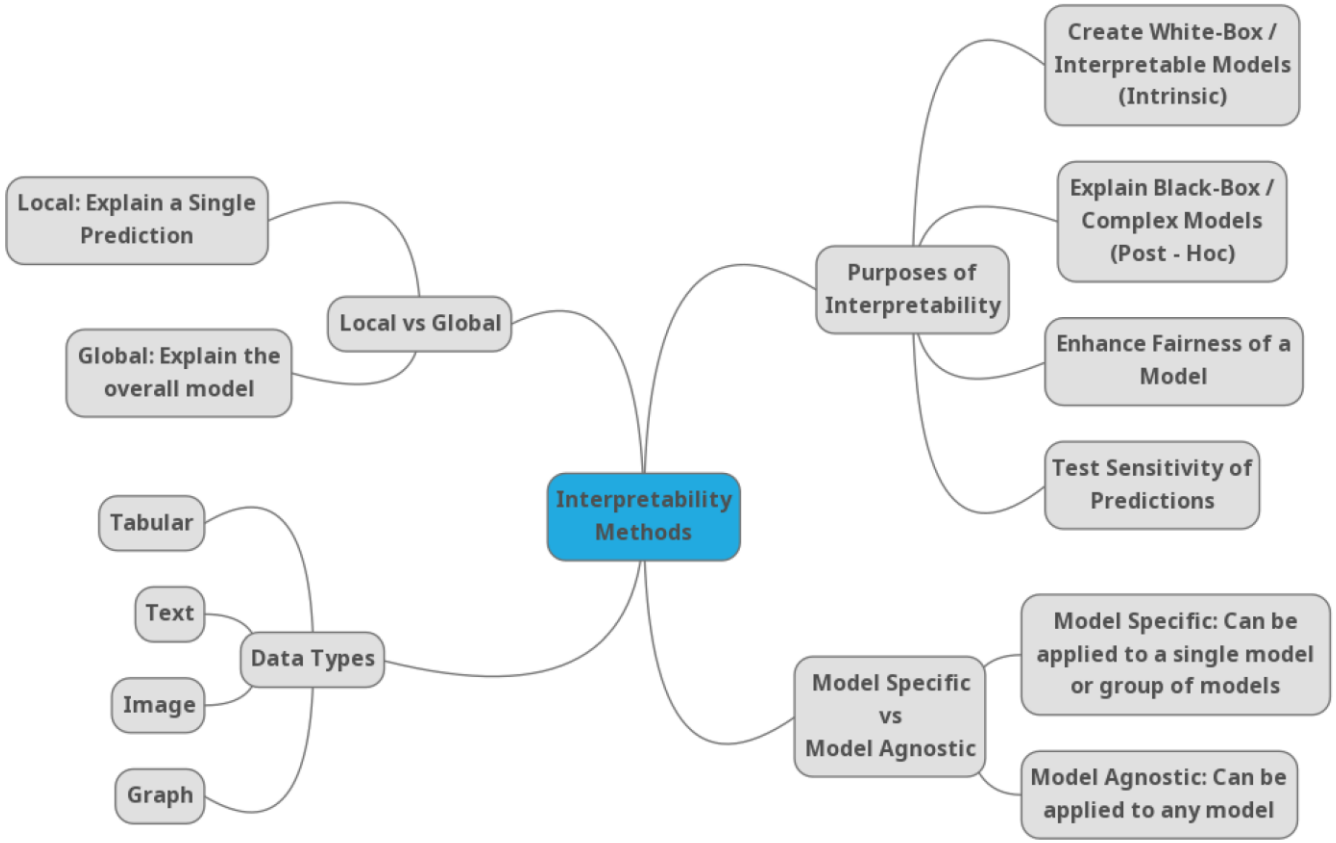
https://renaissancerachel.com/explainable-ai/

Explainable AI



Explainable AI

Interpretability, also often referred to as explainability, in artificial intelligence (AI) refers to the study of how to understand the decisions of machine learning systems, and how to design systems whose decisions are easily understood, or interpretable.



Explainable AI Examples

1. **Feature Importance Analysis**:

   - **Explanation**: Identify which features contribute most to model predictions.

   - **Example**: In a credit risk model, feature importance analysis reveals that credit utilization and payment history are critical factors affecting creditworthiness.

2. **Local Explanations (LIME)**:

   - **Explanation**: Explain individual predictions for specific instances.

   - **Example**: LIME highlights that a loan application was rejected because the applicant's income-to-debt ratio exceeded the threshold.

3. **SHAP (SHapley Additive exPlanations)**:

   - **Explanation**: Assigns a value to each feature's contribution to a prediction.

   - **Example**: SHAP values show that a patient's high cholesterol level significantly influenced the diabetes risk prediction.

Ref – Chat with Co-pilot 28/03/24

Explainable AI

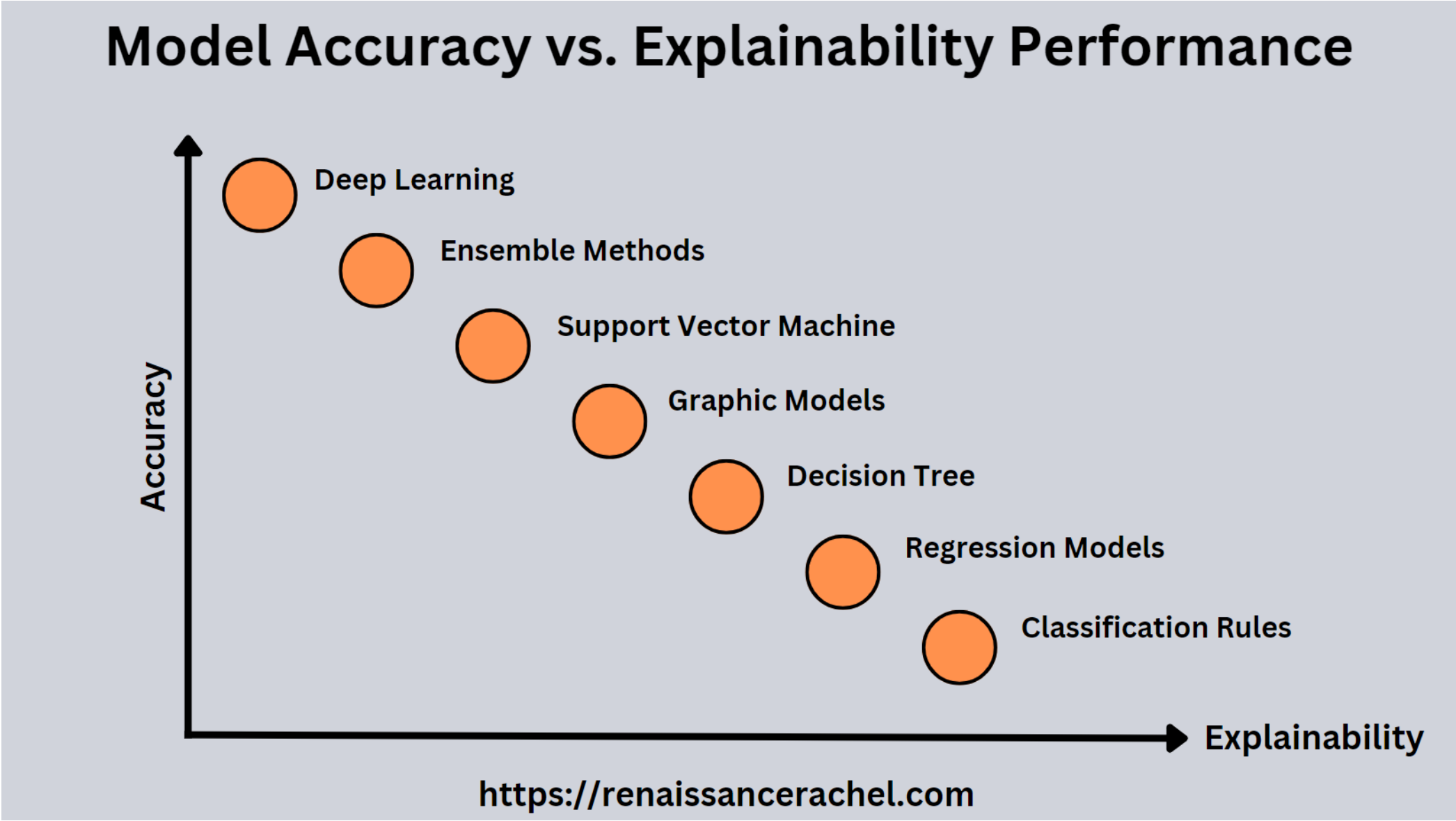
4. **GradCAM (Gradient-weighted Class Activation Mapping)**:

- **Explanation**: Visualizes which parts of an image influenced a deep learning model's decision.
- **Example**: GradCAM highlights the tumor region in a medical image that led to a cancer diagnosis.

5. ****Rule-Based Models**:**

- **Explanation**: Use interpretable models with explicit rules.
- **Example**: A fraud detection system based on decision rules flags transactions with suspicious patterns (e.g., large withdrawals from multiple ATMs in a short time).

Remember, XAI is crucial for building trust and understanding how AI systems arrive at their decisions. 🌟

Ref – Chat with Co-pilot 28/03/24

Explainable AI